PROJECT REPORT | CLOUD SECURITY

Azure Identity and Access Management (IAM) Security Project

Entra ID groups, Azure RBAC and Blob Storage

Platform	Microsoft Azure
Services	Entra ID, Azure RBAC, Blob Storage, Activity Log
Report date	25 October 2026
Prepared by	_____

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1. Executive Summary

This project implements identity and access management in **Microsoft Azure** using **Microsoft Entra ID** security groups and **Azure role-based access control (RBAC)** to protect an **Azure Blob Storage** account, and uses the **Azure Activity Log** to audit management operations. Access is granted to groups rather than individuals so that each job function receives only the permissions it needs.

Three security groups were created (**Azure-Developers**, **Azure-Auditors** and **Azure-Admins**), each with one member user (**Developer**, **Auditor** and **Admin**). Role assignments were then made on the storage account **parikshitarer**: **Contributor** for developers, **Reader** (scoped to a single container) for auditors, and a conditioned **User Access Administrator** role for administrators. The Activity Log confirms that these role assignments, the container creation and related operations all **succeeded**.

The console screenshots show that the roles actually assigned are the broader *management-plane* roles (Contributor, Reader) rather than the planned *data-plane* roles (Storage Blob Data Contributor / Reader). Section 6 compares the plan with what was implemented, and Section 9 gives the recommended corrections. Live access tests, Managed Identities and Key Vault were not completed.

Item	Status
Entra ID security groups	Done – Azure-Developers, Azure-Auditors, Azure-Admins (assigned membership)
Users and group membership	Done – Developer, Auditor and Admin, one group each
Storage account	Done – parikshitarer, East US 2, resource group Development, LRS
RBAC assignments	Done, but roles differ from plan (Contributor / Reader / User Access Administrator)
Anonymous blob access disabled	Planned; setting not visible in the screenshots – verify
Activity Log review	Done – 12 events in the last 6 hours, all Succeeded
Developer / auditor access tests	Not performed or not recorded
Managed Identities, Key Vault	Skipped

2. Objectives and Scope

2.1 Objectives

- Understand Microsoft Entra ID, security groups and Azure RBAC built-in roles.
- Organise access through groups and apply least-privilege permissions to Blob Storage.
- Keep stored data private (no anonymous access).
- Separate developer, auditor and administrator responsibilities.
- Use the Activity Log to investigate who did what, when, and to which resource.

2.2 In scope

- Entra ID groups and users; RBAC role assignments on a storage account and container.
- Review of built-in and privileged administrator roles.
- Activity Log inspection of management-plane events.

2.3 Out of scope (skipped or future)

- Managed identities for workloads and Azure Key Vault.
- Developer-versus-auditor access testing (upload, read, denied-write results).

- Data-plane logging of blob reads and writes, conditional access and MFA policy design.

3. Solution Architecture

Each person is a member of one Entra ID security group. The group, not the person, holds the role assignment, and the assignment is bound to a scope (storage account or container). Management operations such as creating a role assignment are written to the Activity Log.

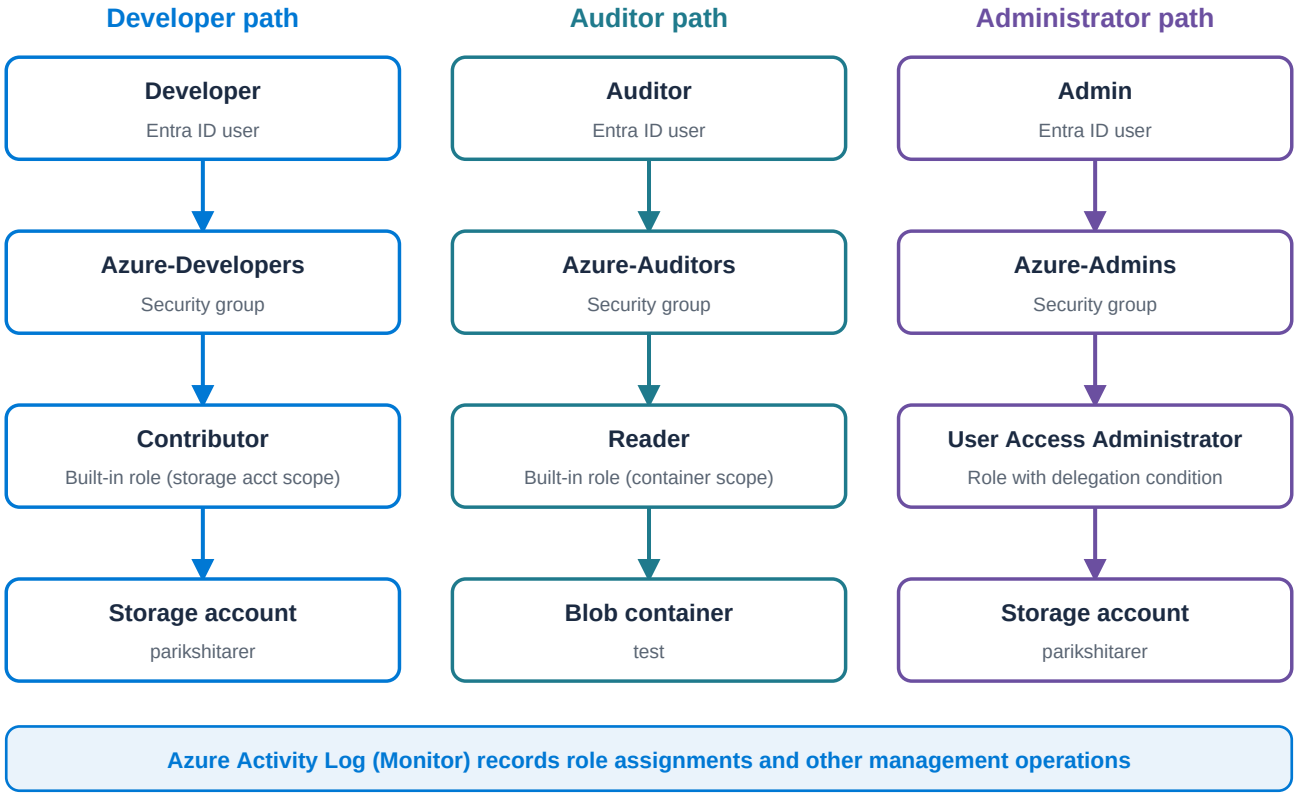


Figure 1 – Access architecture as configured in the console.

4. Azure Services and Resources

Resource	Name / value	Notes
Tenant	Default Directory	Microsoft Entra ID Free license; 0 applications, 0 devices
Security groups	Azure-Developers, Azure-Auditors, Azure-Admins	Type Security, membership Assigned
Users	Developer, Auditor, Admin	Member users in the tenant's onmicrosoft.com domain; account owner is a guest user
Subscription	Azure subscription 1	Scope root for assignments
Resource group	Development	Holds the storage account
Storage account	parikshitarer	East US 2, Standard, LRS, general purpose; hierarchical namespace, SFTP and NFS v3 disabled
Blob container	test	Container targeted by the auditor assignment
RBAC roles	Contributor, Reader, User Access Administrator	Built-in roles
Monitoring	Azure Monitor Activity Log	Management events, 6-hour window reviewed

Note: the project notes use the names Azure-Cloud-Administrators, rg-iam-security and project-data; the console shows **Azure-Admins**, **Development** and **test**. This report uses the names visible in the console.

5. Implementation Steps

Step 1 – Explore Microsoft Entra ID and Azure RBAC

The Default Directory overview shows the tenant, its primary domain and the **Microsoft Entra ID Free** license, with navigation to Users, Groups, Roles and administrators and other identity features. The Free license covers users, groups and RBAC; advanced governance features (for example Privileged Identity Management) require a higher license.

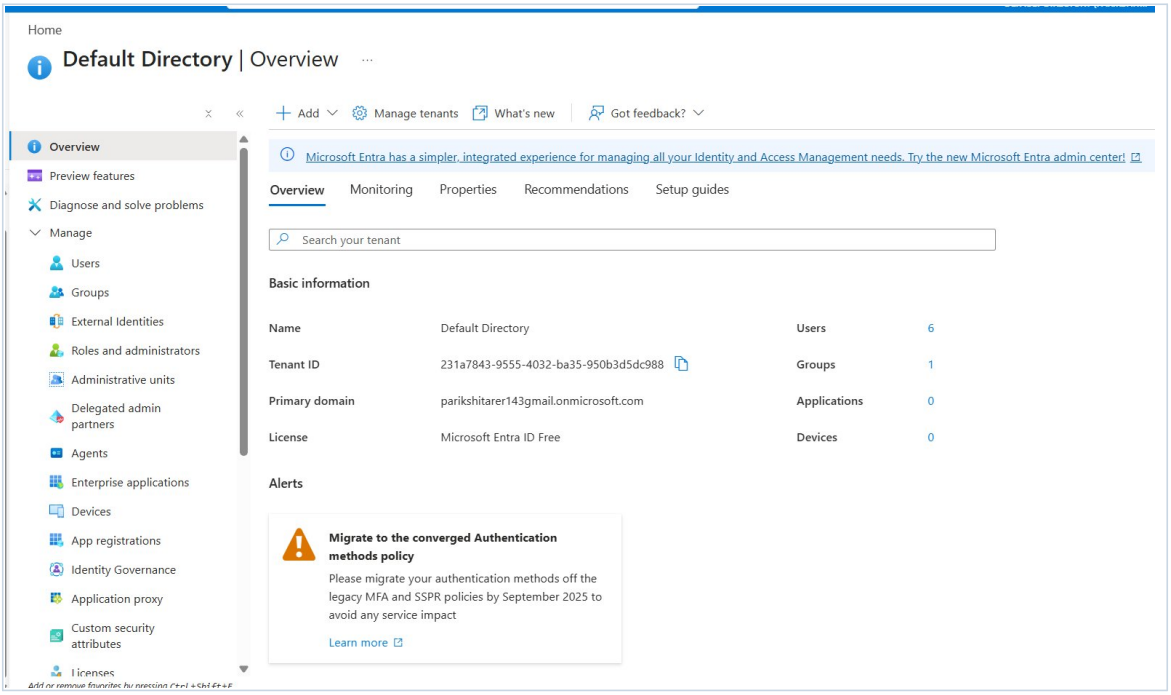


Figure 2 – Microsoft Entra ID: Default Directory overview (Free license).

On the storage account, the Roles tab was used to review the built-in roles (Owner, Contributor, Reader and many service-specific roles) and the **Privileged administrator roles** tab, which lists eight roles that can manage resources or access: Owner, Contributor, Access Review Operator Service Role, Reservations Administrator, Role Based Access Control Administrator, Service Group Administrator, Service Group Contributor and User Access Administrator.

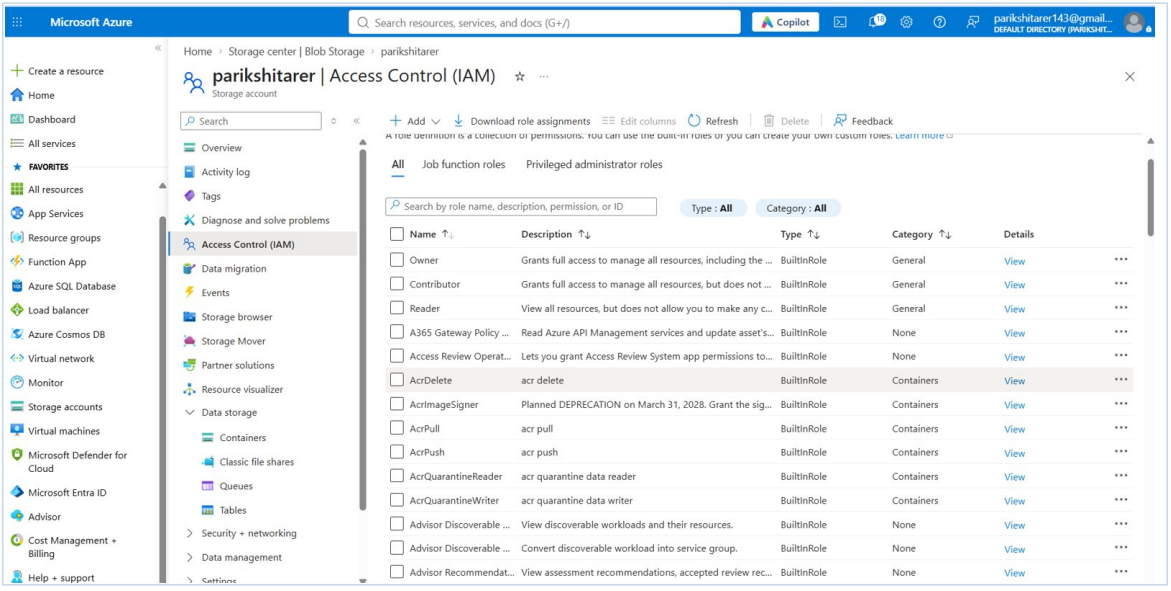


Figure 3 – Built-in roles list on the storage account (All).

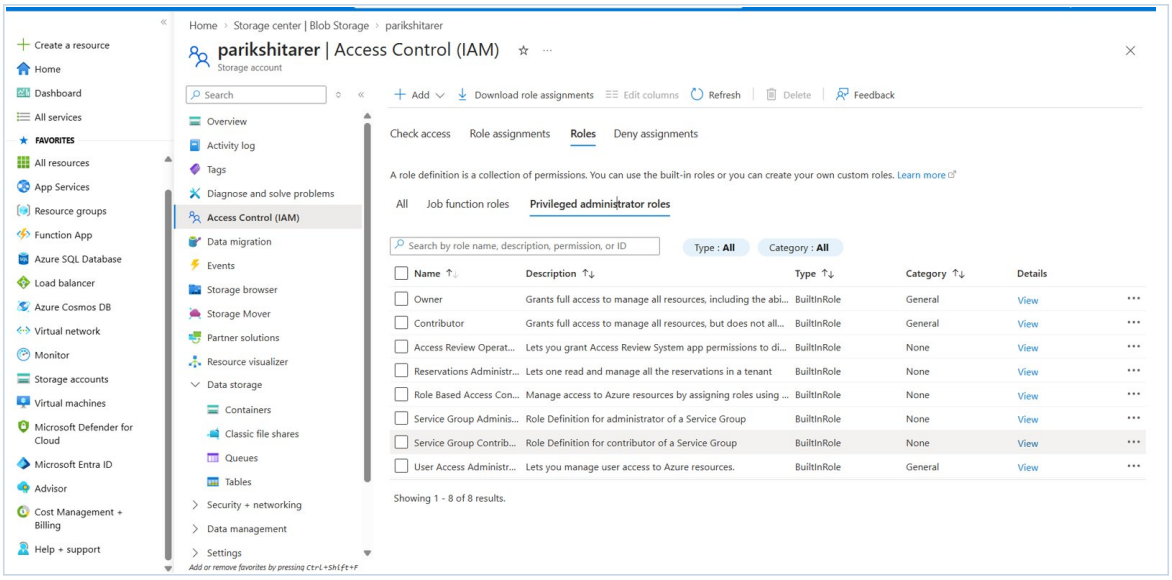


Figure 4 – Privileged administrator roles (8 roles).

Step 2 – Create security groups

Three Entra ID security groups with assigned membership were created. Using groups means that adding or removing a person changes their access in one place, with no per-user role assignments.

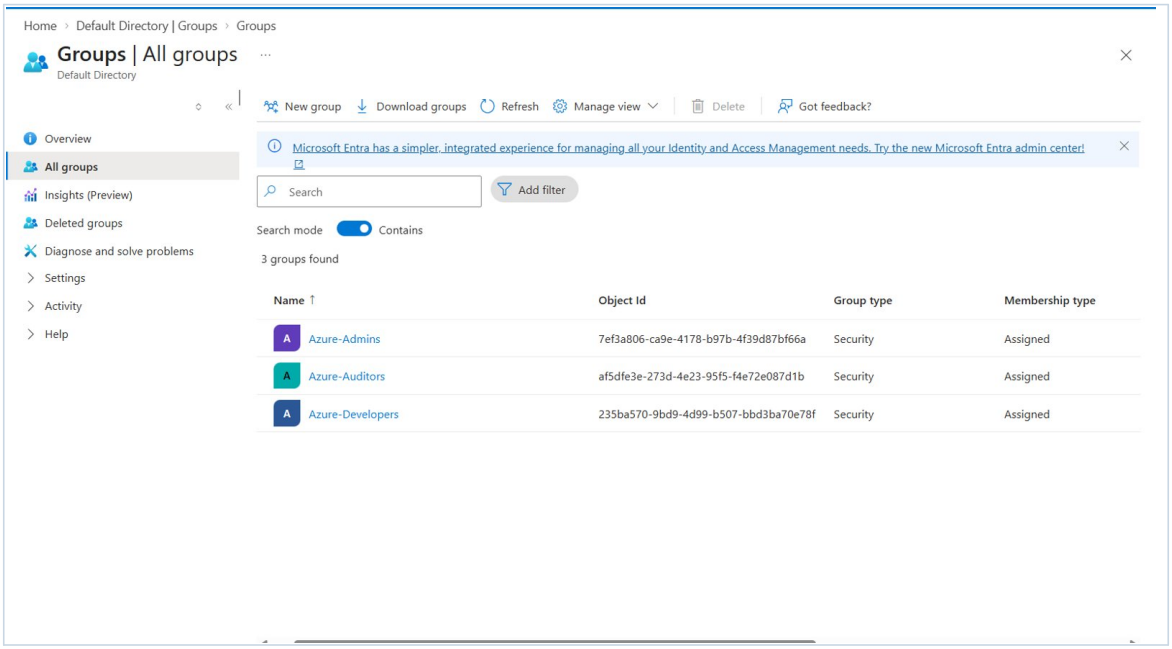


Figure 5 – Entra ID groups: Azure-Admins, Azure-Auditors, Azure-Developers.

Step 3 – Create users and add them to groups

Member users were created and placed into the groups during creation (Assignments tab): **Auditor** into Azure-Auditors, **Developer** into Azure-Developers and **Admin** into Azure-Admins. All accounts were created enabled, with no directly assigned roles at creation.

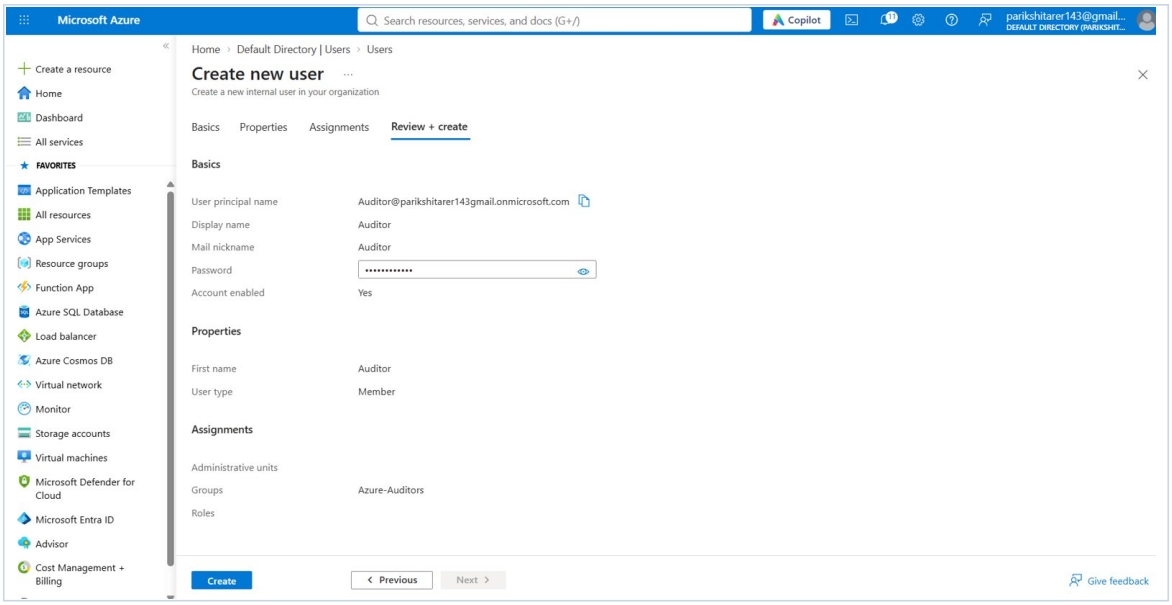


Figure 6 – Create user "Auditor": group Azure-Auditors.

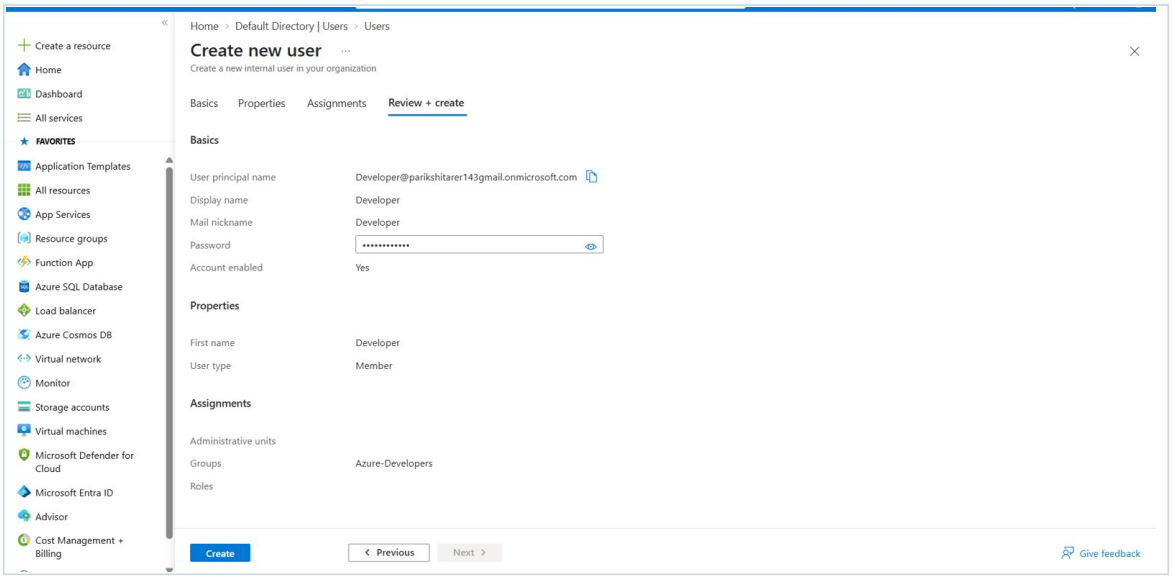


Figure 7 – Create user "Developer": group Azure-Developers.

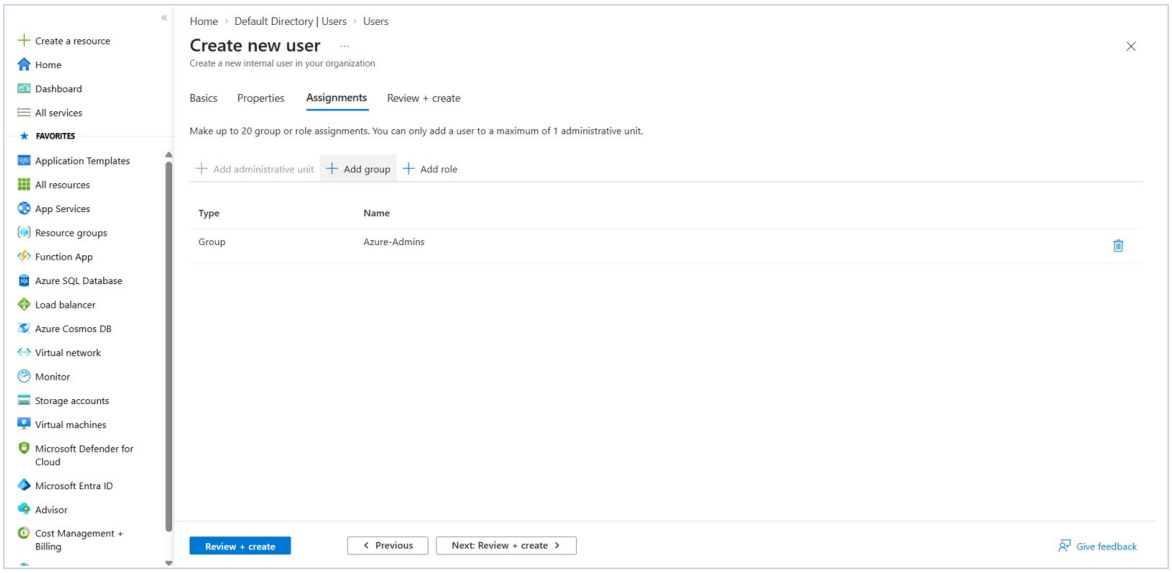


Figure 8 – Assignments tab when creating the Admin user: group Azure-Admins.

Step 4 – Create the storage account

The storage account **parikshitarer** was created in resource group **Development** in **East US 2** with Standard performance and locally redundant storage (LRS). Hierarchical namespace, SFTP, NFS v3 and cross-tenant replication are disabled, which keeps the attack surface small. A private blob container and a test file were then created (the Activity Log shows a *Put blob container* operation). The security settings for secure transfer and anonymous access were not visible in the captured review page and should be confirmed.

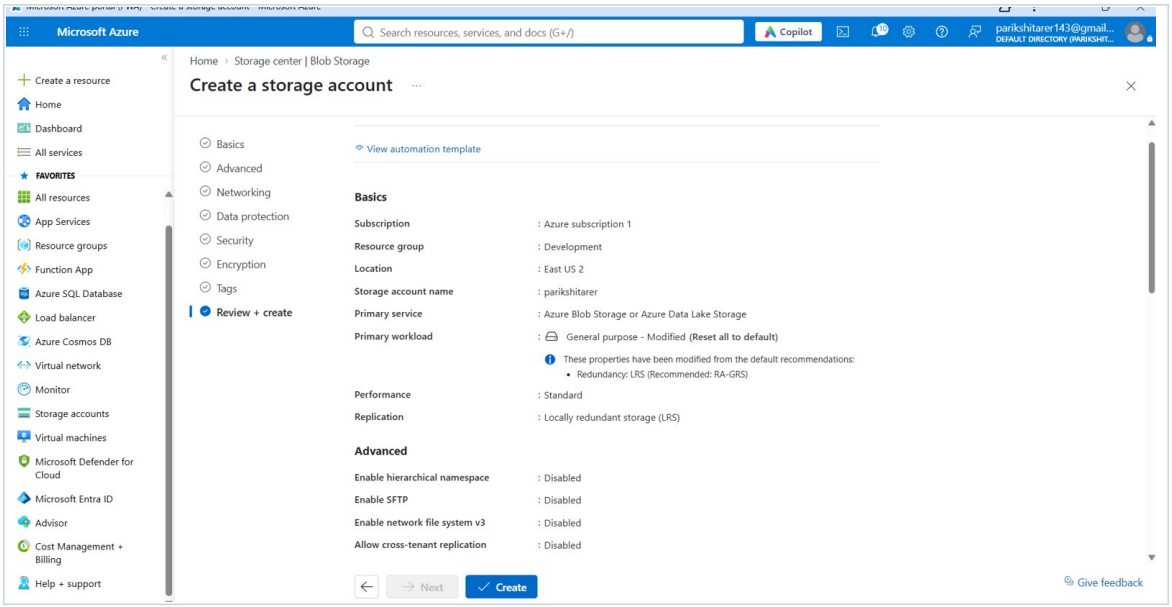


Figure 9 – Create a storage account: review + create summary.

Step 5 – Configure role-based access control

Roles were assigned from **Access Control (IAM)**. The IAM blade offers Check access, Role assignments, Roles and Deny assignments, and the *Add role assignment* wizard (Role, Members, Conditions, Review + assign).

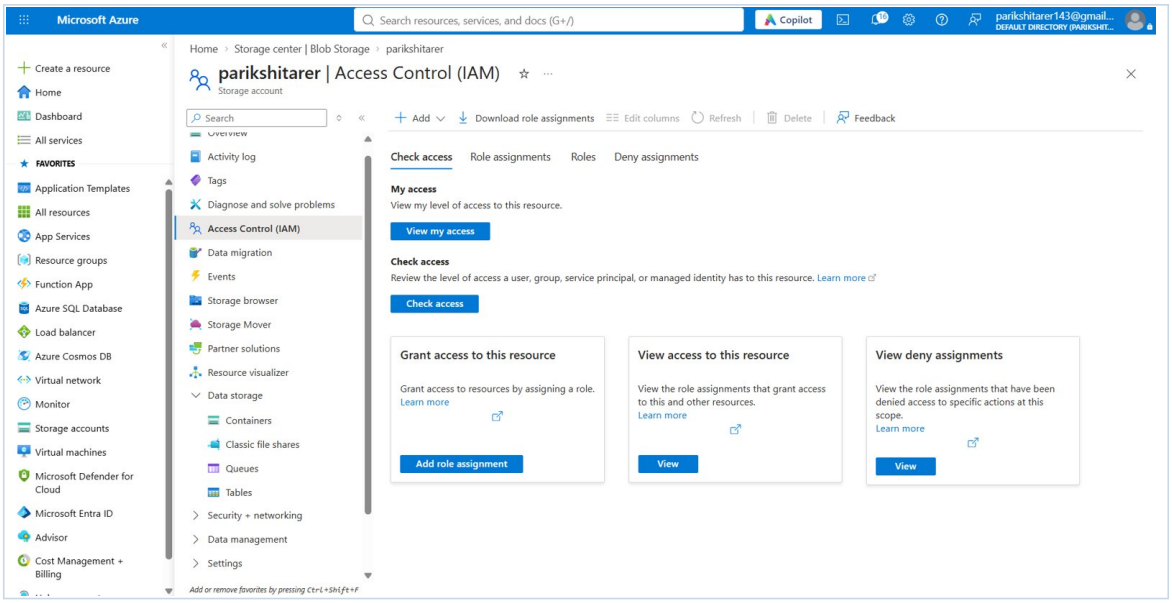


Figure 10 – Storage account Access Control (IAM) blade.

Auditors – Reader at container scope. From the container “test”, the Reader role was assigned to the group Azure-Auditors. The member picker lists the users and groups of the tenant; only the group was selected.

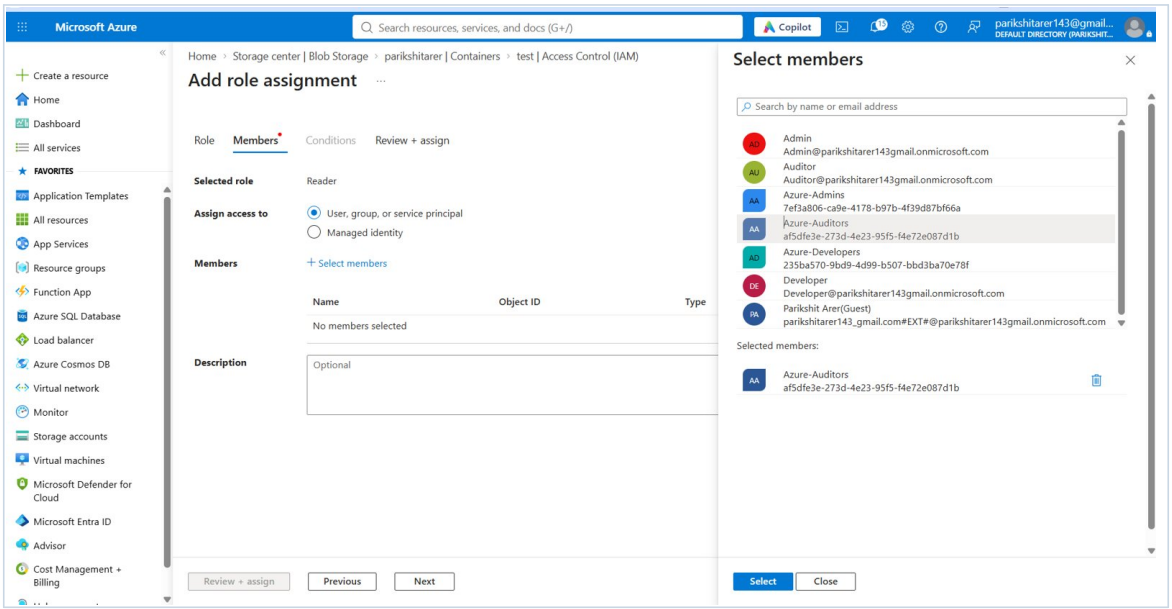


Figure 11 – Add role assignment: selecting the Azure-Auditors group for the Reader role.

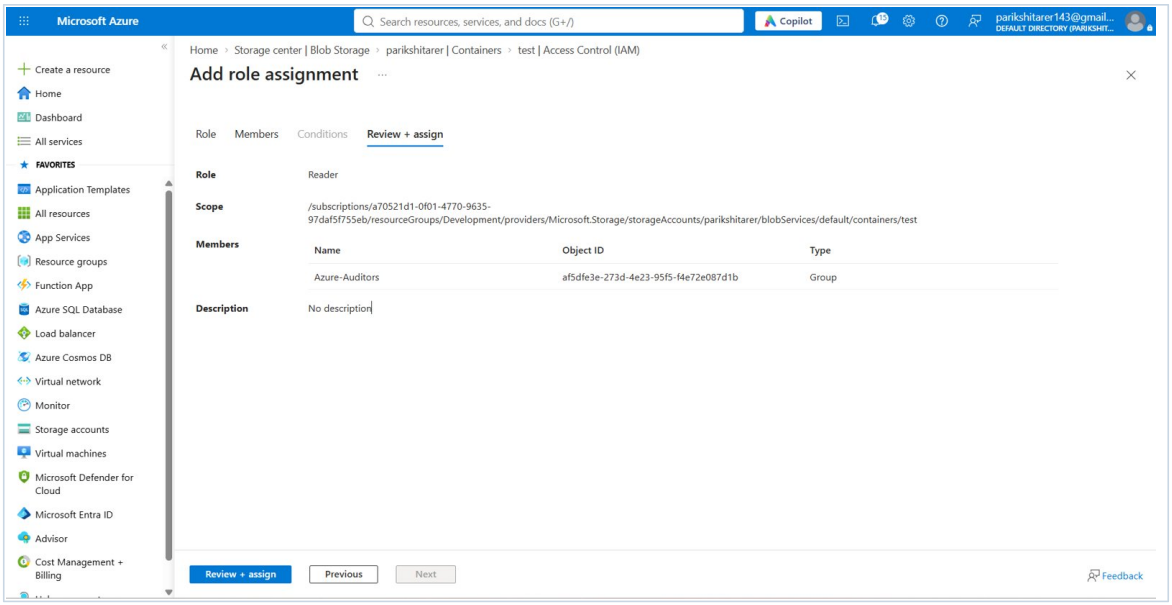


Figure 12 – Review + assign: Reader on container "test" for Azure-Auditors.

Developers – Contributor at storage account scope. The Contributor role was assigned to Azure-Developers on the storage account.

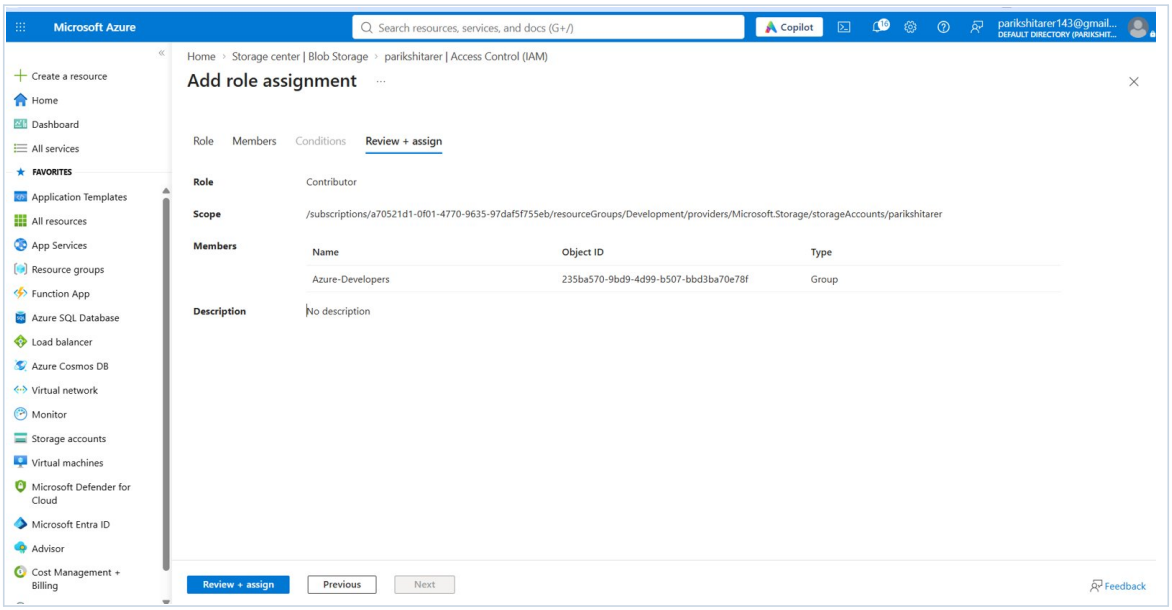


Figure 13 – Review + assign: Contributor on storage account parikshitarer for Azure-Developers.

Administrators – User Access Administrator with a condition. The group Azure-Admins received User Access Administrator on the storage account, restricted by a preconfigured condition that allows assigning all roles *except* Owner, User Access Administrator and Role Based Access Control Administrator. This stops the group from granting itself or others top-level privileges.

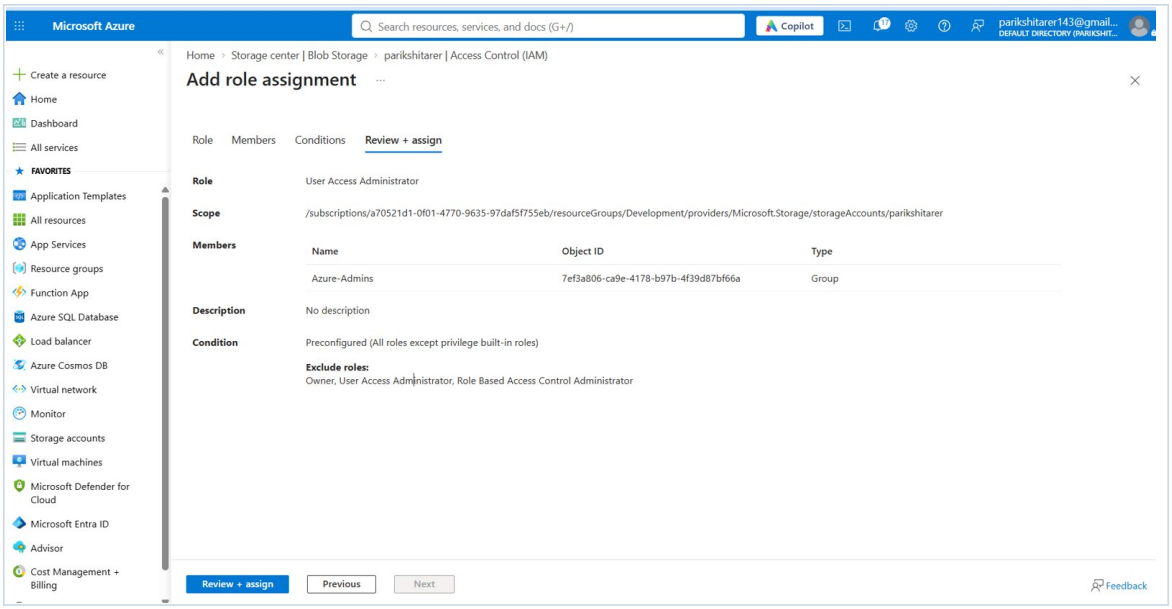


Figure 14 – Review + assign: User Access Administrator with exclusion condition for Azure-Admins.

Step 6 – Entra ID directory role for the Auditor user

The Entra role **Attribute Assignment Reader** (directory scope) lists the user **Auditor** as an assignee. This is a directory role, separate from Azure RBAC, and was assigned to the user directly rather than through the group. It is outside the planned design and is covered in Section 9.

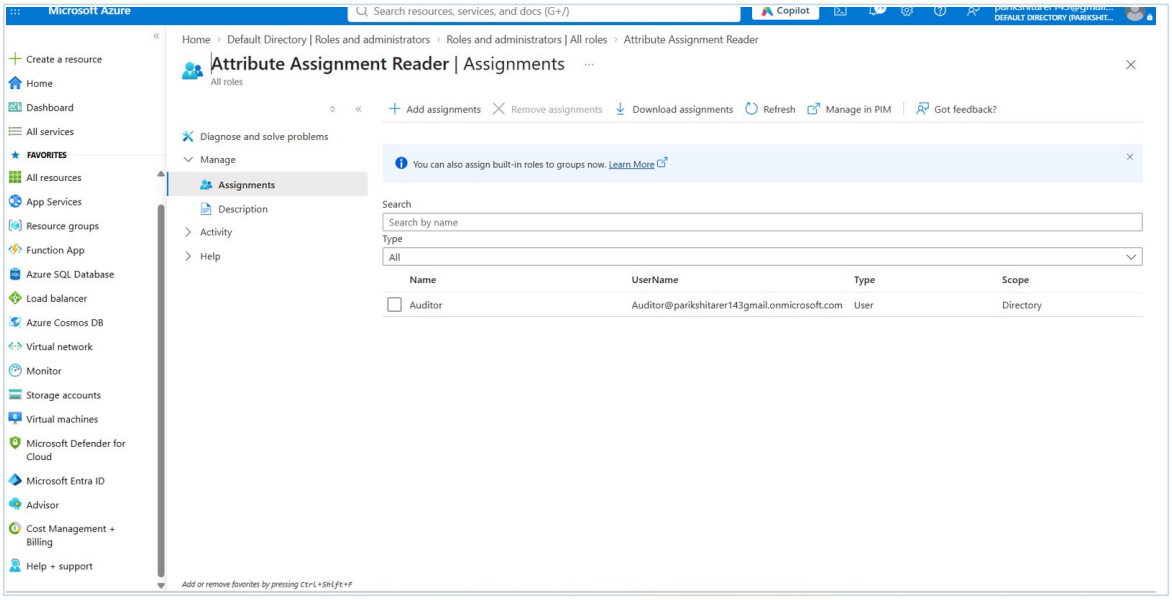


Figure 15 – Entra role Attribute Assignment Reader: assigned to user Auditor (directory scope).

Step 7 – Review the Azure Activity Log

Azure Monitor → Activity log was filtered to Azure subscription 1 for the last 6 hours, returning 12 items, all with status **Succeeded** and initiated by the account owner. The log answers who initiated an operation, its status and timestamp, and the affected resource; expanding an event shows caller, correlation ID and JSON details.

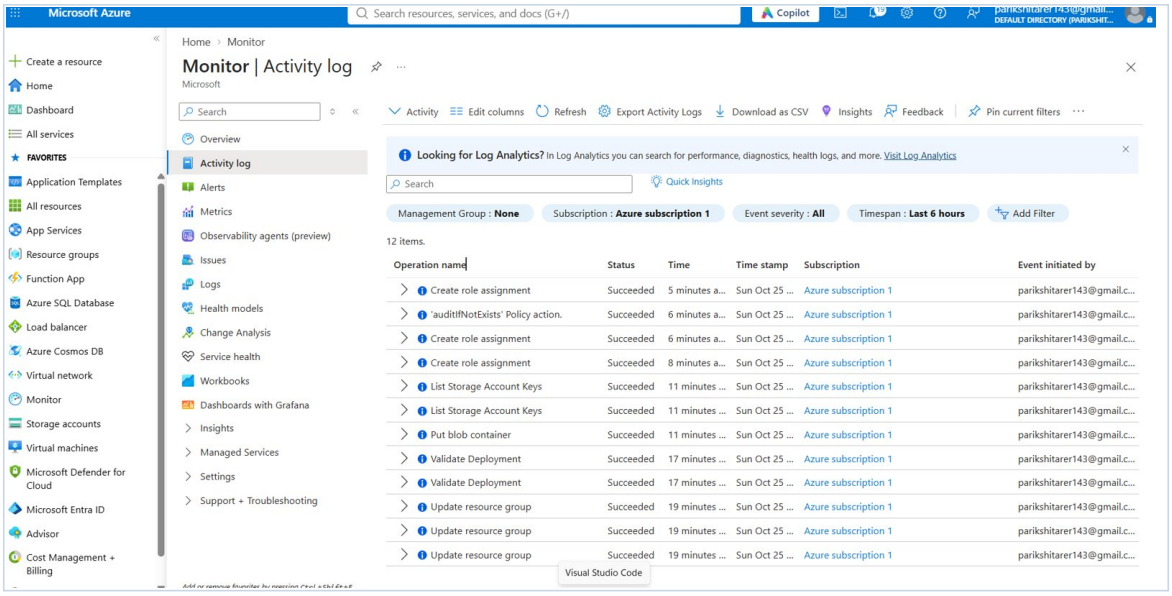


Figure 16 – Monitor | Activity log: 12 management events, all Succeeded.

Operation (oldest to newest)	Events	Meaning
Update resource group	3	Resource group Development created or updated
Validate Deployment	2	Storage account deployment validated
Put blob container	1	Blob container created
List Storage Account Keys	2	Account keys retrieved (typically when the portal browses or uploads blobs using the access key)
Create role assignment	3	The three RBAC assignments: Reader, Contributor and User Access Administrator
‘auditIfNotExists’ Policy action	1	Azure Policy audit evaluation after the change

Important limitation: the Activity Log records *management-plane* operations. It does not automatically record every blob read, upload or delete; for that, enable storage diagnostic settings (data-plane logs) and send them to Log Analytics or a storage account.

6. Planned Design Versus Implemented Configuration

The project notes describe data-plane roles. The console screenshots show different assignments. The table records both so that the report is accurate and the gaps can be closed.

Area	Planned (project notes)	Observed (screenshots)
Developer role	Storage Blob Data Contributor	Contributor on storage account parikshitarer
Auditor role	Storage Blob Data Reader	Reader on container “test”
Administrator group	Azure-Cloud-Administrators; no additional role	Azure-Admins; User Access Administrator (conditioned) on the storage account
Resource group	rg-iam-security (or chosen)	Development
Container	project-data	test
Anonymous access / secure transfer	Disabled / required	Not visible in screenshots
Auditor extra permissions	None	Entra role Attribute Assignment Reader (directory)

Why the role type matters

- **Reader** is a management-plane role: it lets a person see the storage account and container resources, but **not read blob contents**. Blob contents require a data-plane role such as **Storage Blob Data Reader**.
- **Contributor** can manage the whole storage account (change networking and public-access settings, delete it) and can **retrieve the account keys**, which grant full data access and bypass RBAC. It is far broader than a developer needs.
- **Storage Blob Data Contributor** / **Reader** grant only blob data actions, and can be assigned to a single container for least privilege.

7. Security Principles Demonstrated

Principle	How it appears in the project
Group-based access control	Roles are assigned to Azure-Developers, Azure-Auditors and Azure-Admins, not to individual users.
Scoped assignments	The auditor assignment is bound to one container; others to the storage account, not the subscription.
Separation of duties	Different groups for development, auditing and access administration.
Delegation guard-rail	The administrator role carries a condition excluding Owner, User Access Administrator and RBAC Administrator.
Reduced surface	Hierarchical namespace, SFTP and NFS v3 are disabled on the storage account.
Auditability	Activity Log shows the role assignments and resource changes with initiator, status and time.

8. Validation Status

Activity	Status	Evidence
Groups created (3)	Done	Figure 5
Users created and added to groups (3)	Done	Figures 6–8
Storage account created	Done	Figure 9
Role assignments created (3)	Done	Figures 12–14; three Create role assignment events in Figure 16
Activity Log reviewed	Done	Figure 16
Anonymous access disabled; secure transfer required	Not evidenced	Setting not shown
Developer upload / delete test	Not performed	No test results recorded
Auditor read allowed, write denied test	Not performed	No test results recorded
Managed identity, Key Vault labs	Skipped	–

Suggested test plan

- Sign in to the portal as **Developer** (private window). Open the container using *Microsoft Entra user account* as the authentication method; upload `hello.txt`, list and delete it (expected after role correction: allowed).
- Sign in as **Auditor**: list and download blobs (allowed); upload or delete (expected: authorization error).
- As **Admin**: add and remove a test role assignment; confirm that assigning Owner is blocked by the condition.
- Check the Activity Log for the matching events and record the results in a table with screenshots.

9. Findings and Recommendations

#	Observation	Recommendation
1	Auditors have Reader on the container, which does not allow reading blob data.	Assign **Storage Blob Data Reader** to Azure-Auditors at the container scope.
2	Developers have Contributor on the whole storage account, including key retrieval and account management.	Replace with **Storage Blob Data Contributor** at container scope; then disable shared-key access so RBAC is enforced.
3	Azure-Admins holds User Access Administrator, but the notes say no role was to be assigned.	Confirm the intent; keep the exclusion condition and the narrowest scope, or remove the assignment.
4	Auditor has the Entra directory role Attribute Assignment Reader, assigned to the user.	Remove it unless needed; grant roles through groups only.
5	Developer and auditor scopes differ (account vs container); container is “test”, not project-data.	Use one consistent scope and naming scheme for the project container.
6	Anonymous access and secure-transfer settings are not shown.	Verify in Storage account → Configuration: *Allow Blob anonymous access* = Disabled, *Secure transfer required* = Enabled; also set a minimum TLS version.
7	No MFA or conditional access evidence; tenant uses Entra ID Free.	Keep security defaults (MFA) enabled; plan for conditional access and PIM with a higher license.
8	Activity Log shows management events only and keeps a limited history.	Add diagnostic settings: send Activity Log and Blob read/write/delete logs to Log Analytics for long-term audit.
9	Screenshots expose tenant, subscription and object IDs and e-mail addresses.	Blur these values before sharing the report publicly.

10. Labs Skipped and Future Improvements

- **Managed Identities:** no workload identity was configured or tested; the Azure equivalent of an EC2 IAM role.
- **Azure Key Vault:** no vault or secret created; use it with managed identities to remove stored credentials.
- **Access testing:** developer-versus-auditor tests not confirmed complete.
- Further ideas: custom roles, Azure Policy to deny public blob access, access reviews, Infrastructure as Code (Bicep/Terraform).

11. AWS and Azure Comparison

AWS	Azure equivalent
AWS IAM	Microsoft Entra ID and Azure RBAC
IAM users	Microsoft Entra ID users
IAM groups	Microsoft Entra ID security groups
IAM policies and permissions	Azure RBAC roles and assignments
EC2 IAM role	Managed identity assigned to an Azure workload
Amazon S3	Azure Blob Storage
AWS CloudTrail	Azure Activity Log (management events)
AWS Secrets Manager	Azure Key Vault

These are functional comparisons, not exact one-to-one equivalents.

12. Cost Management

Entra ID Free, RBAC and the Activity Log (90-day view) carry no extra charge. Costs come from the storage account (capacity, operations and any diagnostic logs sent to Log Analytics). Delete the test blobs, container and resource group when the lab is finished, and check current Azure pricing for East US 2.

13. Conclusion

The project establishes a working Azure identity foundation: Entra ID security groups with member users, scoped RBAC role assignments on a storage account and container, a guard-railed administrator role, and Activity Log evidence that each change succeeded. The review also shows that the assigned management-plane roles should be replaced with Storage Blob Data roles to meet the least-privilege goal. With that correction, verified access tests, hardened storage settings and data-plane logging, the configuration becomes a solid base for managed identities, Key Vault and automated governance.

Appendix A – Role Assignment Summary

Principal	Role	Scope	Type
Azure-Developers	Contributor	Storage account parikshitarer	Group
Azure-Auditors	Reader	Container test (storage account parikshitarer)	Group
Azure-Admins	User Access Administrator (excludes Owner, UAA, RBAC Admin)	Storage account parikshitarer	Group
Auditor	Attribute Assignment Reader (Entra role)	Directory	User

Appendix B – Recommended Azure CLI Commands

Replace the placeholders with your own IDs. These commands apply the corrections from Section 9 and are provided as a reference; they were not run as part of this project.

```
# Variables
SUB=<subscription-id>; RG=Development; SA=parikshitarer; CT=<container-name>
ACCT="/subscriptions/$SUB/resourceGroups/$RG/providers/Microsoft.Storage/storageAccounts/$SA"
SCOPE="$ACCT/blobServices/default/containers/$CT"

# Developers: blob data read/write/delete on the container only
az role assignment create --assignee-object-id <Azure-Developers-object-id> \
  --assignee-principal-type Group --role "Storage Blob Data Contributor" --scope "$SCOPE"

# Auditors: blob data read-only on the container
az role assignment create --assignee-object-id <Azure-Auditors-object-id> \
  --assignee-principal-type Group --role "Storage Blob Data Reader" --scope "$SCOPE"

# Remove the broad Contributor assignment from developers
az role assignment delete --assignee <Azure-Developers-object-id> --role Contributor \
  --scope "$ACCT"

# Harden the storage account
az storage account update -n $SA -g $RG --allow-blob-public-access false \
  --https-only true --min-tls-version TLS1_2 --allow-shared-key-access false
```

Appendix C – Glossary

Term	Meaning
Microsoft Entra ID	Azure’s cloud identity service (users, groups, applications, sign-in).
Azure RBAC	Role-based access control for Azure resources: a role, a principal and a scope.
Scope	Level at which a role applies: management group, subscription, resource group, resource or container.
Management plane / data plane	Managing the resource itself (create, configure) versus working with the data inside it (read, write blobs).
Storage Blob Data Reader / Contributor	Built-in data-plane roles for reading / reading, writing and deleting blob data.
User Access Administrator	Role that can assign roles to others; can be limited with a condition.
Activity Log	Subscription-level log of management operations with initiator, status and time.
Managed identity	Automatically managed Azure identity for a workload; no stored credentials.